



Radial–axial transport coordination enhances sugar translocation in the phloem vasculature of plants

Mazen Nakad ^{1,*} Jean-Christophe Domec ^{2,3} Sanna Sevanto ⁴ and Gabriel Katul ^{1,3,†}

¹ Department of Civil and Environmental Engineering, Duke University, Durham, North Carolina 27708, USA

² Bordeaux Sciences Agro, UMR 1391 INRA-ISPA, Gradignan 33175, France

³ Nicholas School of the Environment, Duke University, Durham, North Carolina 27708, USA

⁴ Earth and Environmental Sciences Division, Los Alamos National Laboratory, Los Alamos, New Mexico 87545, USA

*Author for correspondence: mazen.nakad@duke.edu

†Senior author

G.K. conceived the original screening and research plans. J.-C.D., S.S., and G.K. supervised the progress of the model development and numerical analysis. M.N. conceived the project and wrote the article with contributions from all the authors, and agreed to serve as the author responsible for contact and communication.

The author responsible for distribution of materials integral to the findings presented in this article in accordance with the policy described in the Instructions for Authors (<https://academic.oup.com/plphys/pages/general-instructions>) is: Mazen Nakad (mazen.nakad@duke.edu).

Abstract

Understanding mass transport of photosynthates in the phloem of plants is necessary for predicting plant carbon allocation, productivity, and responses to water and thermal stress. Several hypotheses about optimization of phloem structure and function and limitations of phloem transport under drought have been proposed and tested with models and anatomical data. However, the true impact of radial water exchange of phloem conduits with their surroundings on mass transport of photosynthates has not been addressed. Here, the physics of the Munch mechanism of sugar transport is re-evaluated to include local variations in viscosity resulting from the radial water exchange in two dimensions (axial and radial) using transient flow simulations. Model results show an increase in radial water exchange due to a decrease in sap viscosity leading to increased sugar front speed and axial mass transport across a wide range of phloem conduit lengths. This increase is around 40% for active loaders (e.g. crops) and around 20% for passive loaders (e.g. trees). Thus, sugar transport operates more efficiently than predicted by previous models that ignore these two effects. A faster front speed leads to higher phloem resiliency under drought because more sugar can be transported with a smaller pressure gradient.

Introduction

The efficiency of photosynthate transport from the production sites (sources; usually leaves) to areas of consumption or storage (sinks) within the vascular tissue known as the phloem is drawing substantial attention in plant physiology. The implications of efficient photosynthate transport range from local impacts on tissue or plant health and growth to ecosystem-scale effects on carbon and water cycling because of the potential link between phloem transport and

stomatal control of photosynthesis (Nikinmaa et al., 2013), and a possible link between phloem transport failure and plant mortality under drought (Sevanto et al., 2014). Consequently, several models for phloem transport and the connection between phloem structure and function, as well as for the potential weak points in the transport system have been formulated (Münch, 1930; Phillips and Dungan, 1993; Thompson and Holbrook, 2003; Jensen et al., 2009, 2012; Sevanto, 2014). The most commonly accepted concept under which all these models operate is that phloem

vasculature is optimized for efficient transport of soluble organic compounds (mostly sugars) produced during photosynthesis approximately as described by the pressure-flow hypothesis or Munch mechanism (Münch, 1930).

In the pressure-flow hypothesis, transport is initiated in leaves when sugars and other metabolic products are loaded into the phloem. Once in the phloem, sugars and water molecules are driven to move through the phloem's complex network of narrow but elongated, interconnected, cylindrical living cells (sieve tubes) spanning the length of the plant following pressure gradients. High sugar concentration at the loading site (leaves) draws water from the xylem, the tissue supplying water to the leaves or other surrounding tissues by osmosis toward the phloem. At the sinks, sugars are unloaded from the phloem to growing or storage cells, and water is released back to the xylem or other surrounding tissues. The loading and unloading at sources and sinks build a pressure gradient in the phloem creating a system where plant water and photosynthate transport over long distances occurs without active pumping. This cycle of pressure buildup and transport without active pumping endowed the phloem system with the label "miracle of ingenuity" (van Bel, 2003).

The simplicity, plausibility, and intuitive appeal of the Munch mechanism have led to its proliferation in mathematical models (Nikinmaa et al., 2013; Jensen et al., 2016). It is routinely used to connect plant carbon sources and sinks, and their concomitant controls in a future CO₂ enriched climate (Mencuccini and Hölttä, 2010; Fatichi et al., 2019). It has been used to explain aspects of plant hydraulic failure during drought (Savage et al., 2017; Huang et al., 2018; Konrad et al., 2018; Sevanto, 2018; Salmon et al., 2019) and extreme cold temperatures (Swanson and Geiger, 1967; Wardlaw, 1968). The direct consequence of those two abiotic stresses should be a decrease in the overall phloem flow rate because the viscosity of a sucrose solution increases substantially with the drought-induced increase in sugar concentration required for osmoregulation (Hölttä et al., 2009) and with decreasing temperature.

However, the Munch mechanism is not free from controversy. The main critique stems from the fact that the sieve tubes seem to have too low of a hydraulic conductance along the phloem to allow sugars to be transported from leaves to roots in the largest and longest of plants (Curtis and Scofield, 1933; Lang, 1979; Fensom, 1981; Knoblauch et al., 2016; Liesche and Schulz, 2018). These studies also report lower leaf sucrose concentration in tall trees compared with shorter vegetation. In general, plants can be categorized based on sucrose loading mechanisms into two groups: passive loading species, that transfer sugars by diffusion from the mesophyll cells into the phloem, and active loading species, that transfer sugars by actively pumping them into the phloem using metabolic energy. Typically, short plants such as herbaceous plants or crops are mainly active sugar loaders and long plants such as trees (Gymnosperm and Angiosperm) exhibit the characteristics of passive sugar

loaders. Supplemental Figure S1 shows that active loaders typically have higher loading concentration (~22% wt/wt) than passive loaders (~15% wt/wt) for all the species considered in each category. This observation seems to contradict the basic assumption of Munch mechanism, because pressure gradients driving water and sugars down the phloem, when formulated as pressure differences between sources and sinks normalized by tube length, must require higher sucrose concentration to compensate for the longer phloem pathway to sustain the same flow velocity.

To resolve the issues of inadequate conductance and too low-pressure gradient to efficiently drive flow in tall plants, it has been suggested that rather than exchanging water and sugars only at the extreme source and sink ends of the phloem pathway, as suggested by the original Munch mechanism, sugars and water could be exchanged at different locations along the pathway essentially forming a "relay" system to facilitate transport (Lang, 1979). While plausible, and, based on modeling studies, also effective in increasing transport capacity (Hölttä et al., 2009), there currently is no clear evidence for unloading and reloading of "sugars" from and to phloem conduits along the transport pathway. There is, however, an increasing amount of evidence that water might readily be exchanged between the conduits and their surroundings along the entire length of the pathway (Knoblauch and Oparka, 2012; Knoblauch and Peters, 2017; Stanfield et al., 2017).

Answering the question of whether and how easily phloem conduits exchange water with their surroundings outside the primary loading and unloading zones at sources and sinks is becoming necessary for evaluating the validity of the Munch mechanism, and because it determines how phloem transport is affected under stress (Sevanto, 2014, 2018). Theoretically, if no water exchange occurs, plants run a risk of blocking phloem flow by viscosity increase and reduced pressure gradient under drought conditions because higher amounts of sugar are needed in the phloem conduits at the loading and unloading zones for osmoregulation against the declining water potential of the xylem and the surrounding tissues. If water exchange occurs readily along the entire transport pathway, the flow may not be restricted by the same constraints that stem from the original interpretation of the Munch mechanism (Phillips and Dungan, 1993; Sevanto, 2014, 2018; Sevanto et al., 2014). In particular, the effects of viscosity buildup can be ameliorated because of the diluting effects of radial water flow, the focus of the work here.

Motivated by these issues, we ask to what degree refinements and addressing the model simplifications in the description of the transport physics within the Munch mechanism enhance the mass flux above and beyond expectations from earlier theories. To address this question generically, an idealized, unsteady, 2-D, osmotically driven pipe flow governed by the physics of the Munch hypothesis is considered. No attempt is made to represent all the complexities of the geometry of the phloem tissue or in the

loading and unloading mechanisms of sugars. Instead, the main difference here stems from the inclusion of the simultaneous effects of concentration-dependent viscosity (Bouchard and Granjean, 1995) where local changes in viscosity with concentration (axially and radially) are allowed. It is shown that including such adjustments to viscous stresses leads to substantial enhancement in the magnitude of the mass flux. Moreover, this enhancement in mass flux is shown to be accompanied by a reduced pressure gradient driving the flow. Thus, the work here adds support to the Munch hypothesis by offering a perspective regarding the contribution of coordination between axial and radial flow to the mass flux. This coordination is enhanced especially in long tubes which explain why high sugar concentration is not needed in tall trees.

Results

The results presented here are generated using numerical simulations for a transient flow with a closed-end tube. The effects of including a concentration-dependent viscosity in the radial and axial directions on the flow are featured. First, the effect of a variable viscosity $\mu(c)$ on the mass flux J is discussed by comparing a 2-D model with variable viscosity and a 2-D model with constant viscosity set by the loading concentration. Second, the enhancement of mass transport due to local coordination between axial and radial movement is presented with reference to long-distance transport in plants.

Transient flow simulations highlight how local variations dynamically affect the overall mass flux especially for closed-ended tube simulation (a limiting case as later discussed in “Governing equations”) where the steady state assumption leads to a uniform concentration everywhere. In the [Supplemental Materials and Methods S1](#), steady state simulations are repeated with a constant initial concentration c_0 at the loading zone and infinite sink at $x/L = 1$ for sugars (all else being the same). Conclusions are not affected by the aforementioned assumptions. In addition, steady state only shows the push of water mass from the tube in contrary to transient simulation that shows the push and pull of water mass.

Moreover, the development of steady state requires a couple of hours based on synthetic experiments (Comtet et al., 2017). In real plants, environmental factors that affect photosynthesis are changing at smaller timescales, and the flux of sucrose that is being loaded into the phloem is likely to change before any development of steady state conditions. However, the findings here about the effects of variable viscosity on mass transport are valid in all these situations independent of whether the flow is at steady state or transient, or the sinks are infinite or finite. These boundary conditions and time scales affect only the magnitude but not the effect itself.

Effect of a concentration-dependent viscosity on $J(c_0)$

Resolving the local changes in viscosity results in an increase in the overall conductivity of the tube above and beyond

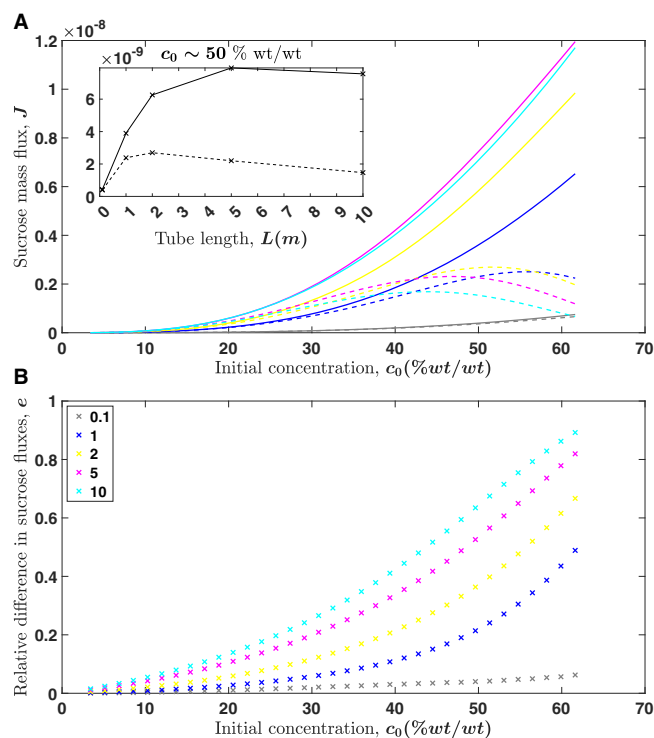


Figure 1 Transient simulation results when the front reaches $x/L = 0.3$ of the domain (see “Model calculations”). A, Sugar mass flux J (kg s^{-1}) as a function of the initial phloem sugar concentration c_0 for the variable viscosity model (solid lines) and the constant viscosity model (dashed lines). Inset plot shows the flux for both models as a function of the tube length when $c_0 \sim 50\%$ wt/wt. B, Relative difference in sucrose fluxes $e = |J_v - J_c| / (J_v + J_c)$ as a function of the initial concentration c_0 . Different tube lengths are presented using different colors.

constant viscosity model (Figure 1A). The variable viscosity model appears to have far higher J than the constant viscosity model at a given c_0 for all tube lengths except for the shortest length ($L = 0.1$ m) where the two models are almost indistinguishable. The effect of the tube length is present in both models where an increase in L leads to an increase in the flux J until a certain value is reached after which J starts to decrease with increasing L . For example, in the variable viscosity model, the value of J when $L = 10$ m is lower than the value of J when $L = 5$ m for the same c_0 as discussed next. An interesting observation is that the value of L for which there is a loss of conductivity (the sugar flux decreased for the same initial concentration c_0) is different for both models, $L = 5$ m for the variable viscosity model and $L = 2$ m for the constant viscosity model. This result is shown in the inset of Figure 1A when $c_0 \sim 50\%$ wt/wt. These calculations were repeated for steady state conditions (constant c_0 at the entrance and infinite sink at exit) for variable L and c_0 in the [Supplemental Materials and Methods S1](#). In all cases, the variable viscosity model yielded higher mass flux compared with the constant viscosity model. However, both models resulted in a reduction in mass flux with increased distance. This can be

attributed to the fact that pull of water in the steady-state model is only active over the entrance of the domain whereas the push is active over the entire L . Hence, it is unsurprising that no optimal L will emerge in the steady-state calculations and the maximum flux will simply decline with increasing L at a given c_0 .

Typical phloem conditions were used to generate the results in both models with constant $\mu(c_0)$ and variable $\mu(c)$: $a = 10 \mu\text{m}$, membrane permeability $k = 5 \times 10^{-14} \text{ m s Pa}^{-1}$ and molecular diffusion coefficient of sucrose in water at standard temperature $D = 4 \times 10^{-10} \text{ m}^2\text{s}^{-1}$. The L was varied from 0.1 to 10 m to describe small or annual plants (e.g. crops) and perennials (trees), respectively. Reflecting the large range of sugar concentration found in plants, the loading sugar concentration inside the tube c_0 was varied from ~ 3.4 to $\sim 61.6\%$ wt/wt, which does not substantially affect the applicability of the Van't Hoff relation and the Newtonian fluid approximation for water. For the constant viscosity model, $\mu(c_0)$ does not vary in space and time. The variable viscosity model solves simultaneously for c and $\mu(c)$ axially and radially in time. Results for larger variation in c_0 (up to $\sim 83.9\%$ wt/wt where the viscosity effects may be overestimated due to the over-estimation introduced by the Van't Hoff relation and the Newtonian fluid approximation used), are presented in the [Supplemental Materials and Methods S2](#).

The importance of variable viscosity can be evaluated by the relative difference in sucrose fluxes

$$e = \left| \frac{J_v - J_c}{J_v + J_c} \right|, \quad (1)$$

where the subscripts “ v ” and “ c ” denote variable viscosity and constant viscosity models, respectively ([Figure 1B](#)). The results show that e increases with L and c_0 as expected. This is because the c_0 affects the overall viscosity value itself and L affects the development of the velocity profile over which viscosity gradients are allowed to build up and increase with increasing L .

2D flow results

The concentration-dependent viscosity leads to an increase in the magnitude of the mass flux J . This is the result of the coordination between axial and radial flow. The direct effect of resolving the viscous stress without any simplification is a smaller viscosity in the vicinity of the membrane since the water influx dilutes the solution near the membrane. This process is also accompanied by an opposite process where the decrease in c near the membrane leads to a lower osmotic effect and radial velocity ([Pedley, 1983](#)). However, in the simulations here, the osmotic forcing is dynamic and dictates the concentration at the membrane through a dynamic boundary condition. This dynamic boundary condition is formulated using Darcy's law (discussed in “Governing equations”) where the effect of an “unstirred” layer ([Pohl et al., 1998](#)) is accounted for. This unstirred layer is further analyzed in the [Supplemental Materials and Methods S3](#) and is shown to be reasonably resolved in the

simulations. The total effect of the aforementioned processes considered here remains an increase in v near the membrane. This increase in radial velocity near the pipe walls, when coupled to a zero radial velocity at the pipe center (as required by symmetry), must be accompanied by an increase in axial velocity gradients (due to the incompressibility approximation). Thus, a speeding up of u is expected.

To understand why the increase in mass flux occurs for the variable viscosity model (vis-à-vis the constant viscosity model), the 2-D simulations are used to illustrate the radial-axial flow and their coordination. These simulations show the local effect of concentration gradients on flow velocity components affected by viscosity and its gradients. Model results show that the computed axial and radial velocities are higher in magnitude because of a lower viscosity after the front ([Figure 2, A–C](#)). Additionally, the pressure gradient driving the flow is lower compared with the constant viscosity case ([Figure 2D](#)). In plants, the concentration gradient between the source and sink leads to viscosity variations in the axial direction. This decrease in viscosity allows a higher radial velocity (out of the phloem) especially near the sink level which increases the efficiency of the flow and leads in return to a higher axial velocity. On the contrary, synthetic experiments ([Jensen et al., 2009](#)) have higher viscosity variations in the radial direction (because of a higher aspect ratio $\epsilon = a/L$ as defined in “Variable viscosity model”) and the increase in radial velocity (into the phloem) is near the source level.

The results presented here are for initial concentration $c_0 \sim 27.4\%$ wt/wt, representing a mean c_0 found across plants, and $L = 2 \text{ m}$ for the time when the sugar front is located at $\sim 30\%$ of the conduit length, chosen for illustration only. The time t it took for the front to reach this location was also different for the two models: $t = 42.62 \text{ s}$ for the variable viscosity model and $t = 51.05 \text{ s}$ for the constant viscosity model (i.e. in the constant viscosity model the flow was ~ 1.2 times slower). Despite this difference in flow velocity, the sugar front at this location appears similar ([Figure 2A](#)). The highest difference is near the front location. The velocity profile, on the other hand, appears to be wider, and the difference between the models is higher after the front position ([Figure 2B](#)). The axial velocity is also higher in the variable viscosity model compared with the constant viscosity model.

The effect of local viscosity gradients generated by concentration gradients is more apparent in the radial velocity than the axial velocity as expected ([Figure 2C](#)). This finding can be anticipated from [Equation \(8\)](#) since the radial velocity profile is directly related to viscosity gradients in the axial direction. These gradients are generated based on concentration gradients from the axial direction that are large, due to the wave nature of the problem (because the advection-diffusion equation has a wave shape especially when it is advection-dominated). The relative difference e between the models is high near the front location. Similar to the axial velocity profile, the radial velocity profile for the variable

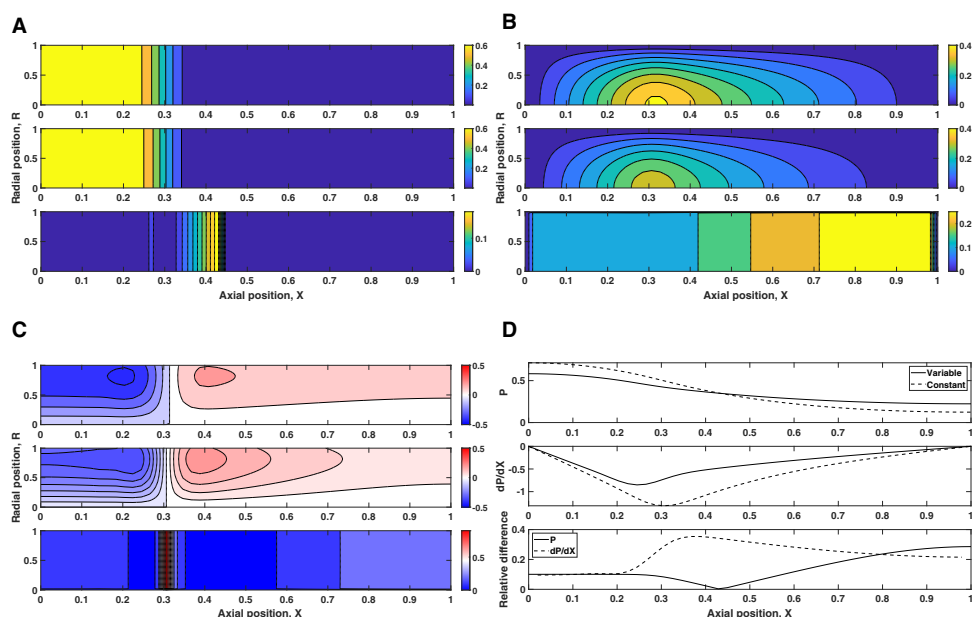


Figure 2 Variations of flow variables as a function of dimensionless axial distance X and radial distance R . Due to cylindrical symmetry, only half of the domain in the radial direction R is shown. Results for a tube length of $L = 2$ m and sugar concentration $c_0 \sim 27.4\%$ wt/wt when the sugar front is located at $\sim 30\%$ of the domain. A, Concentration profiles; B, axial velocity profiles; and C, radial velocity profiles for the variable viscosity model (top), constant viscosity model (middle), and the relative difference (bottom), respectively. D, Pressure (top figure), pressure gradient (middle figure), and their relative difference (bottom figure) for the variable viscosity and constant viscosity models. The relative difference between the models is calculated by e .

viscosity model has a wider velocity range, when compared with the constant viscosity model. Due to a lower sugar concentration near the membrane after the front, the viscosity of the sap decreases leading to less resistance to the radial outflow of water (resulting in a more efficient transport) in the variable viscosity model. This can be conceptually understood as a decrease in wall-friction when area-averaging the equations.

The variable viscosity model also has a smoother pressure field compared with the constant viscosity model (Figure 2D). An interesting result in this figure is that the global pressure gradient over the domain is smaller for the variable viscosity model and yet the front travels at a higher speed. This paradoxical result can only be explained by the increase in conductivity of the tube because of local viscosity effects being coordinated in radial and axial directions.

Discussion

Experimental challenges in measuring water and solute fluxes within the phloem (Curtis and Scofield, 1933; Housley and Fisher, 1977) have led to reliance on mathematical models of simplified phloem transport to understand transport mechanisms in the phloem. As expected when employing such models, values of one or more variables may not be well constrained or are uncertain, possibly by several orders of magnitude. This fact is often used to justify (overly) simplified description of transport physics in models. These simplifications might lead to biased mass fluxes and estimates for total transport. An alternative to the

simplification approach is to assess the effects of the simplification on the results, and in the case of phloem transport, test whether the pressure-flow hypothesis predicts increases or decreases in sugar mass flux when these simplifications are relaxed or re-addressed. In this work, the Munch mechanism is revised to include viscosity variations due to concentration variations in the axial and radial directions within the phloem. Transient flow numerical simulations were conducted to assess the effects of these additions over the conventional approach of setting a single viscosity value that depends on the loading concentration.

These simulations show that the inclusion of viscosity variations with concentration leads to a speed-up in the phloem sap flow when compared with a constant viscosity model. This speed-up is also accompanied with lower overall pressure gradients along the phloem pathway. These co-occurring effects show that viscosity adjustments in a radial–axial coordinated system lead to tube permeability (i.e. conductivity along the phloem pathway) enhancement. In this case, because of increased conductivity, a lower osmotic potential is needed to maintain the same sugar mass flow rate. This finding may offer additional hypotheses about the resilience of plants during drought condition where more negative xylem water tension requires a higher osmotic potential to maintain phloem transport (Sevanto, 2014). For example, if one assumes the xylem water potential can be added as a background pressure (Thompson and Holbrook, 2003), the amount of sucrose needed to overcome this xylem water tension is lower when viscous stress modification

is added to the Munch mechanism. In addition, the sieve elements of the phloem are less prone to failure because of a lower pressure value under the same osmotic potential.

Furthermore, the results here add to the growing evidence on the relation between passive phloem loading and long-distance transport. As discussed in “Introduction”, Gymnosperm and Angiosperm trees are usually passive loaders and there is speculation on the efficiency of the Munch mechanism for these plants. Comtet et al. (2017) showed that these large passive loaders are actually operating at their maximal efficiency using synthetic experiments. Here, numerical simulations show that there is an inverse relation between the size of the plants and the loading concentration for optimally efficient transport. This is shown in Figure 3 where the maximum mass flux ($J/J_{\max} = 1$) exists at an optimal tube length L value that decreases with increasing loading concentration c_0 for both models. However, plant optimal L is expected to drop faster with increasing loading concentration for constant viscosity model than for the variable viscosity model. This finding shows that plants are actually more resilient for increasing phloem length as long as the complexity in transport physics is accommodated.

The increase in the resiliency of tall plants due to a concentration-dependent viscosity is highlighted in the inset of Figure 3, A and B. Due to the lack of data (there are only five tube lengths cases), these figures show the range of concentration where plants operate at maximum efficiency for both

models. They are shown as a sensitivity analysis of the loading concentration for different tube lengths. Consider maize (*Zea mays*) that has typical $c_0 \sim 50\%$ wt/wt and is 2–3 m tall ($= L$) but can grow taller in some cases. Using this c_0 and inspecting Figure 3A, the variable viscosity model predicts efficient transport between 2 and 5 m (because of discrete values in our simulations) whereas the constant viscosity model predicts efficient transport between 1 and 2 m (Figure 3B). This result shows that including viscous stress modification to the Munch hypothesis can confirm the tendencies of some plants to grow taller because of possible competition for light while maintaining the same efficient sugar transport (and keeping high sucrose mass flux as shown in Figure 1).

The importance of using the reformulation of the Munch mechanism to include viscosity variations in a radial–axial coordinated system is mainly affected by the concentration gradient between the source and the sink. When this difference is large, viscosity gradients are large leading to a higher impact. Since this impact is seen as an increased tube conductance, it adds more resilience to losing conductivity when the phloem pathway increases (because of the inverse relation between phloem length and tube conductance). This means that its effect will increase for longer plants.

Materials and methods

To isolate the effect of a concentration-dependent viscosity on radial–axial flow coordination, many simplifications must

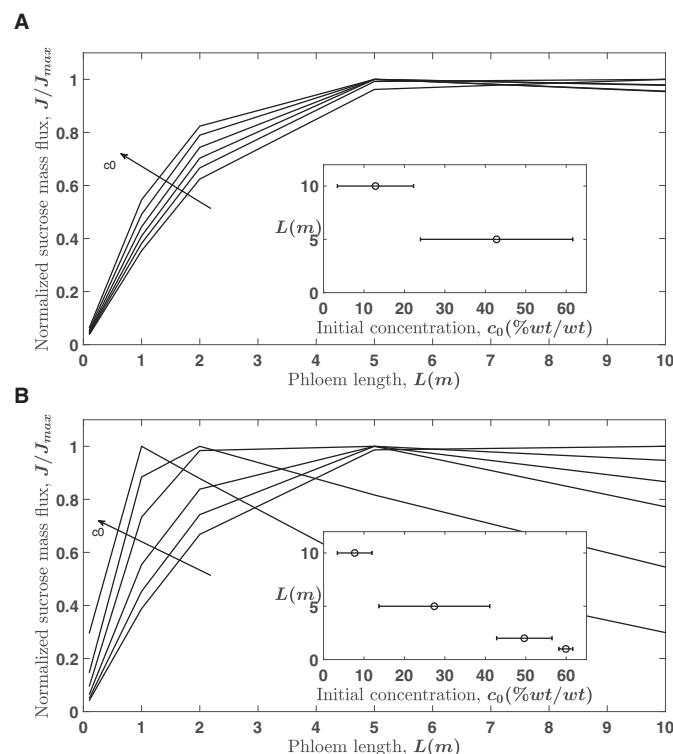


Figure 3 Normalized flux $J/J_{\max}$ as a function of tube length L (m) for the (A) variable viscosity model and (B) constant viscosity model. Direction of the arrow indicates increasing loading concentration c_0 , where $c_0 \sim 10\%$, 20% , 30% , 40% , 50% , and 60% wt/wt. Inset plots show the optimal tube length as a function of the loading concentration for the (A) variable viscosity model and (B) constant viscosity model using a standard error bar type.

still be invoked when representing the physics of translocation in a cylindrical tube. In all formulations considered here, it is assumed that (1) the phloem vasculature can be approximated by a long slender tube of length L and radius a ($\epsilon = a/L \ll 1$) with rigid semipermeable walls characterized by a constant permeability k that allows the exchange of water molecules but not sugars with the surroundings, (2) sieve plates act as dampers to increase wall rigidity and have minimal effect on the flow. They can be modeled as either an “extra” drag force uniformly acting along with L or ignored altogether, (3) the bulk flow is at very low Reynolds number $Re \sim \rho a u \mu^{-1} \ll 1$ where u is a characteristic longitudinal velocity, μ the dynamic viscosity, and ρ the density, so that creeping flow is maintained throughout, and (4) sugar sources and sinks are modeled as boundary conditions at the entry and exit end of the tube. Hence, water can be exchanged with the surroundings but not sugars thereby suppressing any enhancement due to a relay effect.

Variable viscosity model

To derive the general model that includes concentration and viscosity variations in axial and radial directions, the governing equations under certain assumptions and simplifications are first analyzed in Cartesian coordinates. Then, the nondimensional form of these equations, which are necessary for analyzing the numerical model results, is presented in cylindrical coordinates.

The model to be presented in this section is considered one “end-member” case in including a concentration-dependent viscosity. The other “end-member” case this model is compared with is the constant viscosity model that assumes a single viscosity value in the domain set by c_0 . An example of this model for globally averaged steady-state conditions is discussed by (Jensen et al., 2013). Thompson and Holbrook (2003) present a different model that is between these “end-member” cases. In their work, they included local variations in viscosity inside the domain but only used radially averaged equations (i.e. variations in viscosity along radial directions ignored). Due to the nonlinear relation in the viscous stress between velocity and viscosity, area-averaging the equations leads to a simplified model that excludes the effect of this nonlinearity. This issue can be resolved at the expense of solving the equations in axial and radial directions and frames the main approach here.

Governing equations

In a 3-D Cartesian coordinate system defined by longitudinal ($x_1 = x$), lateral ($x_2 = y$), and vertical ($x_3 = z$) directions, water flow within the phloem satisfies the continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_i)}{\partial x_i} = 0, \quad (2)$$

where t is time, $i = 1, 2, 3$ describe direction x_i , and u_i is the instantaneous velocity along direction x_i . Here, index

notation is used with repeated indices implying summation unless otherwise stated. The flow must also satisfy the conservation of momentum, which describes the force balance along direction x_i , and is given as

$$\frac{D(\rho u_i)}{Dt} = \frac{\partial}{\partial x_j} \sigma_{ij}, \quad (3)$$

where $D(\cdot)/Dt$ is the material derivative and σ_{ij} are the nine components of the stress tensor. The σ_{ij} of a Newtonian fluid can be approximated by

$$\sigma_{ij} = -p \delta_{ij} + \mu \left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right), \quad (4)$$

where p and μ are the local pressure and dynamic viscosity of the fluid, respectively, and the δ_{ij} is the Kronecker delta (i.e. $\delta_{ij} = 0$ when $i \neq j$ but unity otherwise). This representation of σ_{ij} is approximate and assumes that the stress tensor is symmetric ($\sigma_{ij} = \sigma_{ji}$) and the so-called second viscosity coefficient (or volume viscosity) is momentarily ignored (Panton, 2006). Equation (4) also assumes that there are no external forces on the fluid and that the gravitational forces cancel out (Thompson and Holbrook, 2003).

In terms of “fluid properties,” ρ depends on c and strictly speaking cannot be treated as constant when c varies in time or along x_i . However, this dependence is minor when compared with variations in μ as demonstrated in the Supplemental Materials and Methods S4 and variations in ρ will be assumed small for simplicity so that $\partial u_i / \partial x_i = 0$. In this case, the σ_{ij} representation given by Equation (4) is reasonable (Panton, 2006). Another common assumption in phloem transport is that μ is constant set by the loading concentration. This approximation is only applicable for small concentration values. However, in plants, c can range from 15% wt/wt to 35% wt/wt (and for maple [Acer] trees even up to $\approx 50\%$ wt/wt). In this high concentration range, the dependence of viscosity on concentration has not been fully analyzed in the context of 3-D water and sugar transport. Some models include this dependence of μ on c in an area-averaged formulation (Thompson and Holbrook, 2003; Jensen et al., 2016), but area-averaged formulations that evolve concentration axially and presume uniform concentration along the radial direction cannot resolve radial-axial flow coordination to be studied here. Therefore, the model proposed here includes the dependence of μ on c in both axial and radial directions and tracks its consequences on the shape of the $J - c$ relation as well as the magnitude of J across differing L and loading concentrations.

In terms of “flow properties,” the low Reynolds number ($Re \ll 1$) and small aspect ratio ($\epsilon \approx 10^{-5}$) can be used to show that under the so-called lubrication theory (where the flow in one of the dimensions is substantially smaller than the others because of geometric constraints), the momentum balance may be approximated by (now written in cylindrical coordinates)

$$\mu \frac{\partial u}{\partial r} = \frac{r}{2} \frac{\partial p}{\partial x}, \quad \frac{\partial p}{\partial r} = 0, \quad (5)$$

where x remains the axial direction with $x = 0$ situated at the loading zone, r is the radial direction with $r = 0$ defining the center of the tube, and $u(x, r)$ and $v(x, r)$ are the axial and radial velocity components, respectively, at any point (x, r) . In [Supplemental Materials and Methods S5](#), the derivation of this formulation and all its assumptions are presented for completeness.

To understand how viscosity gradients affect the flow, [Equation \(5\)](#) can be written in a compact form. First, the tube conductance is re-defined as $K_t = 1/\mu$ as in the [Supplemental Materials and Methods S4](#) (the geometric factor X_f is absent since it is the result of radial averaging). Integrating [Equation \(5\)](#) in the radial direction while noting that the pressure p is only a function of the axial position (as shown in [Supplemental Materials and Methods S5](#)) leads to

$$u(x, r) = \frac{1}{2} \frac{\partial p}{\partial x} \int_0^r K_t(x, r) dr + A(x), \quad (6)$$

where $A(x)$ is an integration function that varies in x . Using the no-slip boundary condition on the longitudinal velocity component $u(x, a) = 0$ at the membrane surface set as $r = a$ leads to $A(x) = -(a^2 < K_t > / 4) (\partial p / \partial x)$, where $< K_t >$ is the radially averaged tube conductance. Similarly, the term $\int_0^r K_t dr$ can be written as $r^2 < K_t >_r / 2$ where the subscript r denotes radial averaging until the current radial position ($r \leq a$). Using this form, [Equation \(6\)](#) describes the axial velocity

$$u(x, r) = \frac{a^2}{4} \frac{\partial p}{\partial x} \left(< K_t >_r \frac{r^2}{a^2} - < K_t > \right), \quad (7)$$

where both terms $< K_t >_r$ and $< K_t >$ are functions of x but only $< K_t >_r$ is a function of r .

[Equation \(7\)](#) is different from the HP expression because the variable viscosity depends on $c(x, r)$ that itself varies radially and axially. To be clear, this dependence is nonlocal because of the integral operator in the r direction. However, if a constant viscosity is used at a given x , $< K_t >_r$ and $< K_t >$ are equal to $1/\mu$, and the aforementioned conservation of momentum equation becomes equivalent to the HP expression with an adjustment. This adjustment is due to osmosis that generates a radial inflow of water leading to $\partial^2 p / \partial x^2 \neq 0$, which then lead to a variable pressure gradient instead of a constant one as is common in HP applications in pipes ([Phillips and Dungan, 1993](#)). However, the partial $\partial p / \partial x$ not being constant does not violate or invalidate the HP equation as discussed elsewhere ([Thompson and Holbrook, 2003](#); [Nakad et al., 2021](#)).

The effect of viscosity gradients is not directly apparent in [Equation \(7\)](#). To make it explicit, the continuity [Equation \(2\)](#) in cylindrical coordinates is now considered. It is given as

$$\frac{\partial u}{\partial x} + \frac{1}{r} \frac{\partial (rv)}{\partial r} = 0. \quad (8)$$

Here, ρ variation with c is once again assumed to be small compared with the viscosity variations with c as stated before. Using the expression for the axial velocity from [Equation \(7\)](#) in the continuity [Equation \(8\)](#), one can see how axial viscosity gradients impact the radial velocity v , which is not identically zero due to osmosis. Moreover, the viscosity gradient is not only the result of the area-averaged tube conductance $< K_t >$ but also stems from the radially averaged (or nonlocal) tube conductance $< K_t >_r$ that depends on radial position r . [Equation \(5\)](#) with [Equation \(8\)](#) can be used to describe the flow of water characterized by $u(x, r)$ (axial velocity) and $v(x, r)$ (radial velocity) inside the tube as a function of position x, r .

[Equations \(5\)](#) and [\(8\)](#), however, remain incomplete since there are two equations with three unknowns u , v , and p . This mathematical setup is in sharp contrast to flow in closed pipes where $v = 0$ everywhere due to the solid wall boundary condition at $r = a$ and symmetry considerations at the pipe center. In phloem, osmosis necessitates a finite v at the pipe walls while symmetry considerations alone result in $v = 0$ at the center of the pipe. Thus, the third equation that relates v to total pressure inside the tube must be provided by osmoregulation. This equation is best formulated as a boundary condition describing a flow through a porous media (a thin membrane here) at $r = a$. Such a boundary condition may be given by a Darcy-type formulation assuming a very low Reynolds number for the radial flow into or out of the pipe walls. This boundary condition yields an expression for v at $r = a$ given by

$$v|_{r=a} = k (p - \Pi|_{r=a}), \quad (9)$$

where $\Pi|_{r=a}$ is the osmotic potential at the membrane and k is the membrane permeability assumed constant and independent of v (i.e. no Forsheimer or quadratic corrections to Darcy's law). This osmotic potential can be related to c using the Van't Hoff relation, $\Pi = R_g T c(x, a)$ as before. This approximation is reasonable for low c and compatible with the assumption that the density and molecular diffusion (discussed below) do not vary appreciably with c when compared with viscosity.

The last equation needed to describe the physics of sugar transport is the conservation of solute mass, which is also needed to solve for u , v , and p . This equation is derived using Reynolds transport theorem that describes the movement of solutes (mainly sugar here) due to advection and molecular diffusion. In cylindrical coordinates, it is given by

$$\frac{\partial c}{\partial t} + \frac{\partial}{\partial x}(uc) + \frac{1}{r} \frac{\partial}{\partial r}(rvc) = D \frac{\partial^2 c}{\partial x^2} + D \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial c}{\partial r} \right), \quad (10)$$

where $D = \nu Sc^{-1}$ is the molecular diffusion coefficient of sugar in water assumed to be again insensitive to c variations when compared with ν , and $Sc \gg 1$ is the molecular Schmidt number for sugars in water (usually of order 10^4).

The final step for describing the physics of sugar transport is to specify the boundary and initial conditions. These are problem-specific and are selected here to illustrate one

restrictive “end member” case of flow in a closed tube with no sugar sinks. This case is dynamically interesting because sugar concentration keeps building up as no sugars are removed from the tube. The other “end-member” case is where sugars are instantly consumed at the end of the pipe (i.e. $c(L, r) = 0$ and sugar sinks are treated as “infinite”). This latter case is expected to lead to a much larger J in the tube, which is why the focus is on the more restrictive former case. In plants, $c(L) = c(0) \ll 1$ and thus osmotic gradients are much higher in the presence of sinks than those set by the closed tube assumption. Thus, the physics of closed tubes must require higher loading concentrations to drive the water velocity, which is why they are more restrictive and thus dynamically interesting from the perspective of exploring limitations on the Munch hypothesis. In a pipe closed at both ends $u(x = 0) = u(x = L) = 0$, water flow must accelerate to a well-defined maximum and then decelerate to zero velocity along L . For initial conditions selected here, sugar is released as an axially smooth function $c(x, t = 0) = f(x)$ with no radial variation, meaning that radial diffusion is initially fast enough to ensure a uniform distribution of sucrose along r . The closed tube assumption with no sinks requires sugar mass to be conserved inside the tube during the entire period resulting in

$$\frac{\partial c}{\partial x}|_{x=0} = \frac{\partial c}{\partial x}|_{x=L} = 0, \quad (11a)$$

$$(vc)|_{r=a} - D \frac{\partial c}{\partial r}|_{r=a} = 0. \quad (11b)$$

Finally, a no-slip boundary condition at the membrane in the axial direction only, that is, $u|_{r=a} = 0$, and symmetry considerations at the center of the tube, where $v|_{r=0} = 0$ and $\partial c / \partial r|_{r=0} = 0$, are all enforced.

Nondimensional form and key dimensionless numbers

To elucidate the key dimensionless numbers governing water and sugar movement, and make interpretation of the equations easier, this section describes the scaling analyses and the nondimensional form of the equations used. To write the equations in dimensionless form, the following scales were adopted: The x and r were scaled by the tube length L and radius a , respectively, leading to $x = L X$, $r = a R$. Time, and axial and radial velocity as well as pressure and concentration were scaled by their respective initial values at $x = 0$ (subscript 0) leading to $t = t_0 \tau$, $u = u_0 U$, $v = v_0 V$, $p = p_0 P$, $c = c_0 C$, and $\mu = \mu_0 \sim \mu$. The dynamic viscosity μ was scaled by μ_0 , determined from c_0 and T . This μ_0 will also be used in a constant viscosity model as a reference to assess the impact of accommodating variable viscosity in σ_{ij} and its spatial gradients. Using these scales, the nondimensional equations for the two velocity components and pressure are

$$\frac{\partial U}{\partial X} + \frac{1}{R} \frac{\partial RV}{\partial R} = 0, \quad (12a)$$

$$\tilde{\mu} \frac{\partial U}{\partial R} = \frac{R}{2} \frac{\partial P}{\partial X}, \quad (12b)$$

$$V|_{R=1} = M P - C|_{R=1}, \quad (12c)$$

where $u_0 = k R_g T c_0 \epsilon^{-1}$, $v_0 = \epsilon u_0$, $p_0 = \mu_0 L u_0 a^{-2}$, and $M = k \mu_0 L^2 a^{-3}$ is the Munch number defined as the ratio of axial resistance over radial resistance as discussed in Jensen et al. (2009) and Nakad et al. (2021). The complete scaling analysis for the Navier–Stokes equations is shown in Supplemental Materials and Methods S5.

The nondimensional form of the conservation of sugar mass, that is, Equation (10), is

$$\frac{\partial C}{\partial \tau} + Pe \frac{\partial}{\partial X} (UC) + Pe \frac{1}{R} \frac{\partial}{\partial R} (RVC) = \epsilon^2 \frac{\partial^2 C}{\partial X^2} + \frac{1}{R} \frac{\partial}{\partial R} \left(R \frac{\partial C}{\partial R} \right), \quad (13)$$

where $t_0 = a^2 D^{-1}$ is the radial diffusion timescale and $Pe = v_0 a D^{-1}$ is the radial Peclet number defined by the ratio of radial advection to radial diffusion. This expression explicitly shows the relative contributions of radial flow dynamics (through Pe) and simplified geometry (through the slender ratio ϵ) to mass transport.

Model calculations

The proposed model calculations provide the variations of $J(x, r)$, $c(x, r)$, $u(x, r)$, $v(x, r)$, and $p(x, r)$ at every t . A description highlighting the numerical method used to obtain the results is presented in Supplemental Materials and Methods S6. To link a representative J with a c_0 in a manner that allows comparison with the prior $J_{\max}$ and optimal c analysis, the following steps and approximations were taken in post-processing the model results. With a radial Peclet number $Pe \ll 1$, it is reasonable to assume that the mass flow primarily occurs in the axial direction (Nakad et al., 2021). The molecular diffusion can also be neglected since the axial Peclet number, defined by the ratio of axial advection to axial diffusion and derived from Pe by $Pe_l = \epsilon^{-2} Pe$, is large (i.e. $Pe_l \gg 1$). Using these assumptions, the area-averaged sugar flux can be reasonably determined from post-processing time variations of the sugar front position x_f . This front is also delineated from maximal $|\partial c(x, r) / \partial x|$. To determine x_f , we fitted an exponential relation between x_f and t using (Jensen et al., 2009; Nakad et al., 2021)

$$x_f = L - (L - l) \exp\left(-\frac{t}{\tilde{t}}\right), \quad (14)$$

where $l \approx 0.2$ is the initial sucrose front position, $L = 1$ is the length of the tube and x_f is the front position all in nondimensional form. Subtracting l from both sides of Equation (14), a linear relation between t and $\ln[(X_{\text{final}} - x_f) / X_{\text{final}}]$ (where $X_{\text{final}} = L - l \approx 0.8$ and

$X_f = x_f - l$) can be obtained. Hence, linear regression applied to the 2D numerical solution was then used to obtain the constant $\hat{t}$ away from the entrance boundary condition (when the front reaches around 30% of the domain). The front speed was determined as $U_s = dx_f/dt$. Finally, the mass flux was approximated by $J_{\text{num}} = Q_{\text{num}} C$, where Q_{num} is the numerical volumetric flow rate. The Q_{num} was determined using approximated speed $Q_{\text{num}} = A_t U_s$, where A_t is the cross-sectional area of the tube. We present results from the 2-D model simulation for the axial velocity U , radial velocity V , concentration C , and pressure P in the dimensionless form to illustrate the effect of variable viscosity on radial and axial variations of these variables. The model simulations were conducted using MATLAB programming language (Mathworks, Natick, MA, USA).

Supplemental data

The following materials are available in the online version of this article.

Supplemental Figure S1. Histogram showing the reported number of species (ordinate) operating at the measured phloem sugar concentration (c_0) values (abscissa) taken from Jensen et al. (2013).

Supplemental Figure S2. Steady-state simulation results for a tube length of $L = 2$ m and sugar concentration $c_0 \sim 27.4\%$ wt/wt with constant c_0 at the domain entrance ($X = 0$) and infinite sink at the exit ($X = 1$).

Supplemental Figure S3. Sugar mass flux J (kg s^{-1}) as a function of the initial phloem sugar concentration c_0 for the variable viscosity model (solid lines) and the constant viscosity model (dashed lines) using the steady-state numerical simulation.

Supplemental Figure S4. Transient simulation results for a sugar concentration $c_0 \sim 68.5\%$ wt/wt and tube length $L = 2$ m when the sugar front is located at $\sim 25\%$ of the domain ($x_f \approx 0.25$).

Supplemental Figure S5. Normalized sugar mass flux (J/J_{max}) as a function of the initial (or loading) concentration c_0 for the constant viscosity model with $\nu = \nu(c_0)$ in a closed tube and transient simulation.

Supplemental Materials and Methods S1. Steady-state simulation.

Supplemental Materials and Methods S2. Extrapolation to extreme sugar concentration.

Supplemental Materials and Methods S3. Thickness of the unstirred layer.

Supplemental Materials and Methods S4. Optimal sugar concentration.

Supplemental Materials and Methods S5. Model derivation.

Supplemental Materials and Methods S6. Numerical scheme.

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